of the founding fathers of computer science. He was also fluent in seven languages. His colleagues at Princeton said, only half-joking, that von Neumann could prove anything—and anything he proved was correct.

Von Neumann published his proof as part of his textbook on quantum physics in 1932. There's no evidence that Einstein was even aware of this proof, but many other physicists were—and for them, merely the idea of a proof from the mighty von Neumann was enough to settle the debate. The philosopher Paul Feyerabend experienced this firsthand after attending a public talk given by Bohr: "At the end of the lecture [Bohr] left, and the discussion proceeded without him. Some speakers attacked his qualitative arguments—there seemed to be lots of loopholes. The Bohrians did not clarify the arguments; they mentioned the alleged proof by von Neumann and that settled the matter… like magic, the mere name of 'von Neumann' and the mere word 'proof' silenced the objectors."

At least one person did notice a problem with von Neumann's proof shortly after it was published. Grete Hermann, a German mathematician and philosopher, published a paper in 1935 criticizing von Neumann's proof. Hermann pointed out that von Neumann failed to justify a crucial step, and thus the whole proof was flawed. But nobody listened to her, partly because she was an outsider to the physics community—and partly because she was a woman.

Despite the flaw in von Neumann's proof, the Copenhagen interpretation remained totally dominant. Einstein was painted as an old man out of touch with the rest of the world, and questioning the Copenhagen interpretation became tantamount to questioning the massive success of quantum physics itself. And so quantum physics continued for the next twenty years, piling success upon success, without any further questions about the hole at its heart.

Why does quantum physics need an interpretation? Why doesn't it simply tell us what the world is like? Why was there any dispute between Einstein and Bohr at all? Einstein and Bohr certainly agreed that quantum physics worked. If they both believed the theory, how could they disagree about